```
clc;
clear;
close all;
```

1. INPUT PARAMETERS

```
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

A = 0.9;
f = 50;
Fs = 1000;
N = 200;
B = 8;
SF = 0.5;

n = 0:N-1;
x = A*sin(2*pi*f*n/Fs);

% Q1.7 fixed-point format
S = 2^(B-1);
Delta = 1/S;
Qmin = -1;
Qmax = 1-Delta;
```

2. IDEAL FILTER RESPONSE

```
y = filter(b,a,x);
```

3. FIXED-POINT QUANTIZATION

```
ys = round(y*S)/S;
```

4. OVERFLOW DETECTION

```
ov = (y < Qmin) | (y > Qmax);

No_Overflow = sum(ov);
Peak_Unscaled = max(abs(y));
```

5. SATURATION

```
y_clipped = min(max(y,Qmin),Qmax);
ysat = round(y_clipped*S)/S;
```

6. WRAP-AROUND

```
yint = round(y*S);
ywrap_int = mod(yint+S,2*S)-S;
ywrap = ywrap_int/S;
```

7. SIGNAL SCALING

```
xs = SF*x;

y_scaled_ideal = filter(b,a,xs);
yscale = round(y_scaled_ideal*S)/S;

Peak_Scaled = max(abs(yscale));

No_Overflow_Scaled = sum( ...
    (y_scaled_ideal < Qmin) | (y_scaled_ideal > Qmax));
```

8. ERROR CALCULATIONS

```
RMSE_Sat = sqrt(mean((y-ysat).^2));

RMSE_Wrap = sqrt(mean((y-ywrap).^2));

RMSE_Scale = sqrt(mean((y_scaled_ideal-yscale).^2));
```

9. DISPLAY RESULTS

```
fprintf('\n=======================================\n');
fprintf(' FIXED-POINT ARITHMETIC RESULTS\n');
fprintf('=======================================\n');

fprintf('Word Length = %d bits\n',B);
fprintf('Fixed-Point Format = Q1.7\n');
fprintf('Quantization Step = %.7f\n',Delta);
fprintf('Representable Range = [%.7f, %.7f]\n',Qmin,Qmax);

fprintf('\nUnscaled Peak = %.6f\n',Peak_Unscaled);
fprintf('Unscaled Overflow Count = %d\n',No_Overflow);

fprintf('\nSaturation Peak = %.6f\n',max(abs(ysat)));
fprintf('Wrap-around Peak = %.6f\n',max(abs(ywrap)));

fprintf('\nScaling Factor = %.2f\n',SF);
fprintf('Scaled Peak = %.6f\n',Peak_Scaled);
fprintf('Scaled Overflow Count = %d\n',No_Overflow_Scaled);

fprintf('\nSaturation RMSE = %.6e\n',RMSE_Sat);
fprintf('Wrap-around RMSE = %.6e\n',RMSE_Wrap);
fprintf('Scaled Output RMSE = %.6e\n',RMSE_Scale);
```

```
=======================================
 FIXED-POINT ARITHMETIC RESULTS
=======================================
Word Length = 8 bits
Fixed-Point Format = Q1.7
Quantization Step = 0.0078125
Representable Range = [-1.0000000, 0.9921875]

Unscaled Peak = 0.901600
Unscaled Overflow Count = 0

Saturation Peak = 0.898438
Wrap-around Peak = 0.898438

Scaling Factor = 0.50
Scaled Peak = 0.453125
Scaled Overflow Count = 0

Saturation RMSE = 2.389472e-03
Wrap-around RMSE = 2.389472e-03
Scaled Output RMSE = 2.219846e-03
```

10. OUTPUT TABLE

```
Method = {
    'Unscaled'
    'Saturation'
    'Wrap-around'
    'Scaled'
};

Peak = [
    Peak_Unscaled
    max(abs(ysat))
    max(abs(ywrap))
    Peak_Scaled
];

RMSE = [
    0
    RMSE_Sat
    RMSE_Wrap
    RMSE_Scale
];

Overflow = [
    No_Overflow
    sum((y < Qmin) | (y > Qmax))
    sum((y < Qmin) | (y > Qmax))
    No_Overflow_Scaled
];
```

```
T = table(Method,Peak,RMSE,Overflow);

fprintf('\n=======================================\n');
fprintf(' OUTPUT PARAMETER TABLE\n');
fprintf('=======================================\n');

disp(T);
```

```
=======================================
 OUTPUT PARAMETER TABLE
=======================================
        Method           Peak        RMSE       Overflow
    _______________    _______    _________    ________

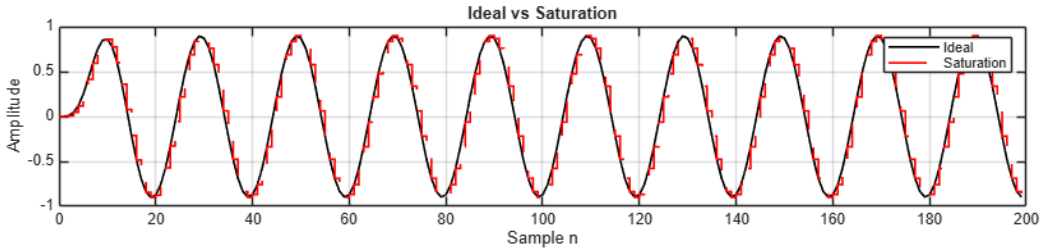
    {'Unscaled'   }     0.9016            0        0
    {'Saturation' }    0.89844    0.0023895        0
    {'Wrap-around'}    0.89844    0.0023895        0
    {'Scaled'     }    0.45312    0.0022198        0
```

11. GRAPH 1: IDEAL VS SATURATION

```
figure;

plot(n,y,'k','LineWidth',1.2);
hold on;
stairs(n,ysat,'r--','LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Ideal vs Saturation');
legend('Ideal','Saturation');
```



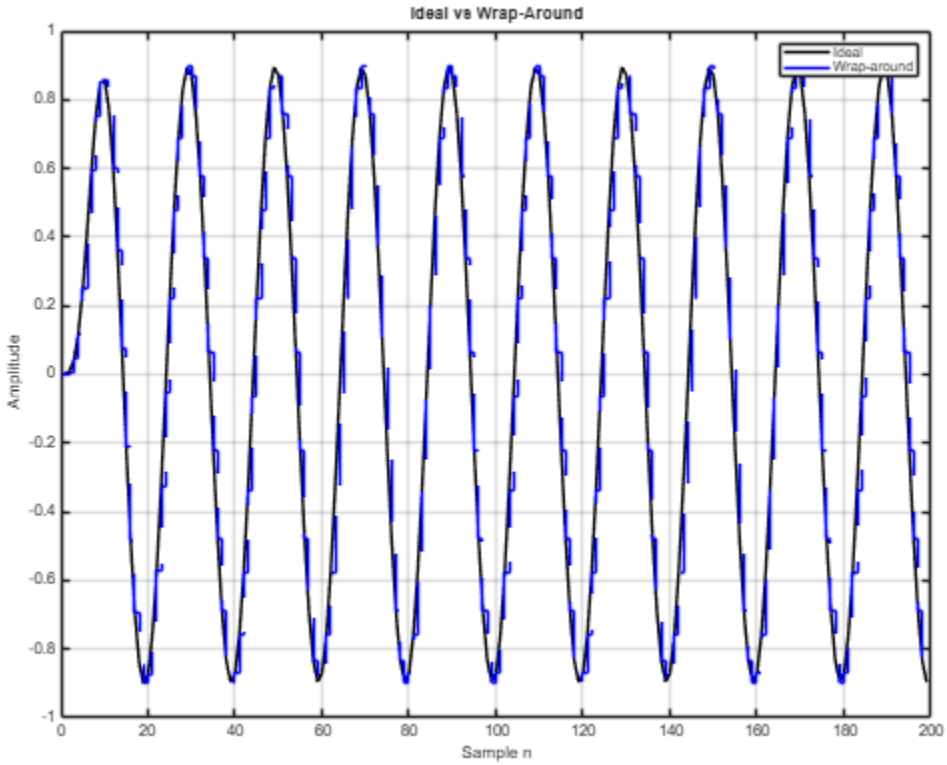
12. GRAPH 2: IDEAL VS WRAP-AROUND

```
figure;

plot(n,y,'k','LineWidth',1.2);
hold on;
stairs(n,ywrap,'b--','LineWidth',1.2);

grid on;
```

```
xlabel('Sample n');
ylabel('Amplitude');
title('Ideal vs Wrap-Around');
legend('Ideal','Wrap-around');
```

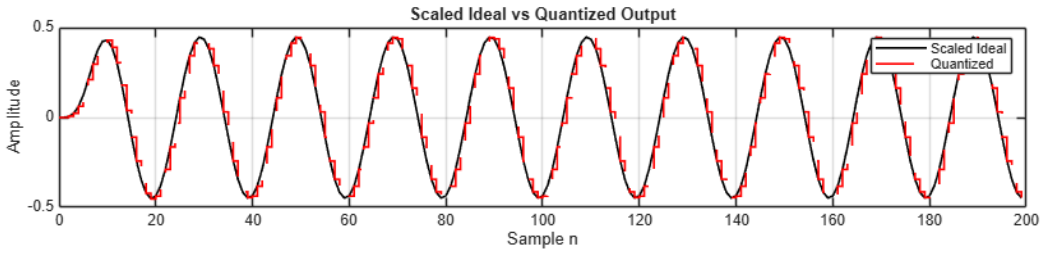


13. GRAPH 3: SCALED OUTPUT

```
figure;

plot(n,y_scaled_ideal,'k','LineWidth',1.2);
hold on;
stairs(n,yscale,'r--','LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Scaled Ideal vs Quantized Output');
legend('Scaled Ideal','Quantized');
```

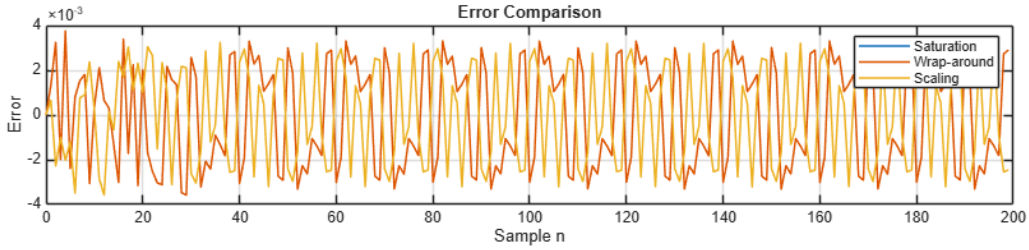


14. GRAPH 4: ERROR COMPARISON

```
figure;

plot(n,y-ysat,'LineWidth',1.1);
hold on;
plot(n,y-ywrap,'LineWidth',1.1);
plot(n,y_scaled_ideal-yscale,'LineWidth',1.1);

grid on;
xlabel('Sample n');
ylabel('Error');
title('Error Comparison');
legend('Saturation','Wrap-around','Scaling');
```



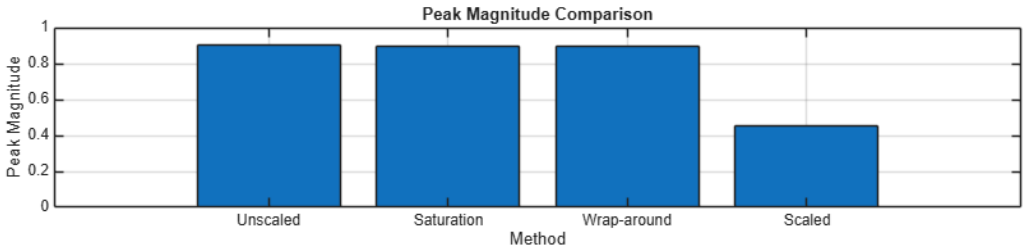
15. GRAPH 5: PEAK MAGNITUDE COMPARISON

```
figure;

bar(Peak);

grid on;
set(gca,'XTick',1:4);
set(gca,'XTickLabel',Method);

xlabel('Method');
ylabel('Peak Magnitude');
title('Peak Magnitude Comparison');
```



16. FINAL RESULT

```
fprintf('\n=======================================\n');
fprintf(' FINAL RESULT\n');
fprintf('=======================================\n');

fprintf('1. Fixed-point quantization completed.\n');
fprintf('2. Overflow detection completed.\n');
fprintf('3. Saturation and wrap-around evaluated.\n');
fprintf('4. Signal scaling evaluated.\n');
fprintf('5. Quantization errors calculated.\n');
fprintf('6. All five graphs generated.\n');
```

```
=======================================
 FINAL RESULT
=======================================
1. Fixed-point quantization completed.
2. Overflow detection completed.
3. Saturation and wrap-around evaluated.
4. Signal scaling evaluated.
5. Quantization errors calculated.
6. All five graphs generated.
```

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